

MODEL QUESTION PAPER 2

**S4 INSTRUMENTATION ENGINEERING**

***SUBJECT: PROCESS CONTROL INSTRUMENTATION***

*Time: 3 Hour*

*Max. Marks: 75*

***I: Part –A: Answer all the following questions in one word or one sentence. (1 x 9 = 9Marks)***

|   |                                                                 | <i>Module Outcomes</i> | <i>Cognitive Level</i> |
|---|-----------------------------------------------------------------|------------------------|------------------------|
| 1 | Define manipulated variable.                                    | M1.02                  | U                      |
| 2 | Define process.                                                 | M1.01                  | R                      |
| 3 | Name two types of discontinuous controller modes.               | M2.02                  | U                      |
| 4 | Write one advantage of PD controller                            | M2.03                  | U                      |
| 5 | Write the equation for PID controller.                          | M2.03                  | R                      |
| 6 | Write the equation to calculate rangability of a control valve. | M3.04                  | U                      |
| 7 | List any one application of globe valve.                        | M3.03                  | U                      |
| 8 | Define alarm.                                                   | M4.01                  | U                      |
| 9 | What is FSK?                                                    | M4.03                  | U                      |

***II: Part – B: Answer any eight questions. Each question carries 3 Marks. (3 x 8 = 24 Marks)***

|     |                                                              | <i>Module Outcomes</i> | <i>Cognitive Level</i> |
|-----|--------------------------------------------------------------|------------------------|------------------------|
| 1.  | Explain self-regulation of a process.                        | M1.03                  | U                      |
| 2.  | Describe variable range of a controller.                     | M1.03                  | U                      |
| 3.  | Describe neutral zone in a discontinuous controller mode.    | M2.02                  | U                      |
| 4.  | Describe integral controller.                                | M2.03                  | U                      |
| 5.  | Define control valve coefficient (CV).                       | M3.02                  | U                      |
| 6.  | Draw neat diagram of pneumatic actuator.                     | M3.04                  | U                      |
| 7.  | Describe the working of limit switch.                        | M3.04                  | U                      |
| 8.  | Explain the advantages of Profibus DP.                       | M4.03                  | U                      |
| 9.  | Draw the block diagram of HART digital communication system. | M4.04                  | U                      |
| 10. | Write short notes on two position controller mode.           | M2.02                  | U                      |

**Part – C**

**Answer all questions. Each question carries 7 marks. (7 x 6 = 42 Marks)**

|      |                                                                             | <i>Module Outcomes</i> | <i>Cognitive Level</i> |
|------|-----------------------------------------------------------------------------|------------------------|------------------------|
| III  | Explain process equation with figure.<br>OR                                 | M1.03                  | U                      |
| IV   | Describe flow control system with example.                                  | M1.04                  | U                      |
| V    | Explain electronic error detector with figure.<br>OR                        | M2.03                  | U                      |
| VI   | Describe electronic Proportional integral (PI) controller mode with figure. | M2.03                  | U                      |
| VII  | Describe I to P converter.<br>OR                                            | M3.04                  | U                      |
| VIII | Illustrate the inherent flow characteristics of control valve.              | M3.03                  | U                      |
| IX   | Explain motion balanced current telemetry system.<br>OR                     | M4.02                  | U                      |
| X    | Illustrate the role of alarms in process control with an example.           | M4.01                  | A                      |
| XI   | Describe position telemetry system with figure.<br>OR                       | M4.02                  | U                      |
| XII  | Explain basic HART specification.                                           | M4.04                  | U                      |
| XIII | Describe pressure control system with example.<br>OR                        | M1.04                  | U                      |
| XIV  | Describe temperature control system with example.                           | M1.04                  | U                      |

Question Wise Analysis

Course: **PROCESS CONTROL INSTRUMENTATION** – Question -2

| <b>Q. No</b> | <b>Module Outcome</b> | <b>Cognitive Level</b> | <b>Marks</b> | <b>Time(Min.)</b> |
|--------------|-----------------------|------------------------|--------------|-------------------|
| I.1          | M1.02                 | U                      | 1            | 2                 |
| I.2          | M1.01                 | R                      | 1            | 2                 |
| I.3          | M2.02                 | U                      | 1            | 2                 |
| I.4          | M2.03                 | U                      | 1            | 2                 |
| I.5          | M2.03                 | R                      | 1            | 2                 |
| I.6          | M3.04                 | U                      | 1            | 2                 |
| I.7          | M3.03                 | U                      | 1            | 2                 |
| I.8          | M4.01                 | U                      | 1            | 2                 |
| I.9          | M4.03                 | U                      | 1            | 2                 |
| II.1         | M1.03                 | U                      | 3            | 8                 |
| II.2         | M1.03                 | U                      | 3            | 8                 |
| II.3         | M2.02                 | U                      | 3            | 8                 |
| II.4         | M2.03                 | U                      | 3            | 8                 |
| II.5         | M3.02                 | U                      | 3            | 8                 |
| II.6         | M3.04                 | U                      | 3            | 8                 |
| II.7         | M3.04                 | U                      | 3            | 8                 |
| II.8         | M4.03                 | U                      | 3            | 8                 |
| II.9         | M4.04                 | U                      | 3            | 8                 |
| II.10        | M2.02                 | U                      | 3            | 8                 |
| III          | M1.03                 | U                      | 7            | 16                |
| IV           | M1.04                 | U                      | 7            | 16                |
| V            | M2.03                 | U                      | 7            | 16                |
| VI           | M2.03                 | U                      | 7            | 16                |
| VII          | M3.04                 | U                      | 7            | 16                |
| VIII         | M3.03                 | U                      | 7            | 16                |
| IX           | M4.02                 | U                      | 7            | 16                |
| X            | M4.01                 | A                      | 7            | 16                |
| XI           | M4.02                 | U                      | 7            | 16                |
| XII          | M4.04                 | U                      | 7            | 16                |

|      |       |   |   |    |
|------|-------|---|---|----|
| XIII | M1.04 | U | 7 | 16 |
| XIV  | M1.04 | U | 7 | 16 |

Course : **PROCESS CONTROL INSTRUMENTATION –QP 2**

Blue Print

Mark Distribution

| Module | hr / Module | Marks/Module<br>( $h_i / \sum H_i$ ) * 123 ( $\pm 5\%$ ) | Type of Questions |       |                 |       |                 |       |                 |       |
|--------|-------------|----------------------------------------------------------|-------------------|-------|-----------------|-------|-----------------|-------|-----------------|-------|
|        |             |                                                          | Part A            |       | Part B          |       | Part C          |       | Total           |       |
|        |             |                                                          | No of Questions   | Marks | No of Questions | Marks | No of Questions | Marks | No of Questions | Marks |
| 1      | 15          | $31 \pm 6$                                               | 2                 | 2     | 2               | 6     | 4               | 28    | 6               | 36    |
| 2      | 15          | $31 \pm 6$                                               | 3                 | 3     | 3               | 9     | 2               | 14    | 10              | 26    |
| 3      | 14          | $29 \pm 6$                                               | 2                 | 2     | 3               | 9     | 2               | 14    | 9               | 25    |
| 4      | 16          | $33 \pm 6$                                               | 2                 | 2     | 2               | 6     | 4               | 28    | 6               | 36    |
| Total  | 60          | 123                                                      | 9                 | 9     | 10              | 30    | 12              | 84    | 31              | 123   |

Blue Print

Cognitive Level Mark Distribution

| Cognitive Level | Marks | % of Marks |
|-----------------|-------|------------|
| Remembering     | 2     | 1.5        |
| Understanding   | 114   | 93         |
| Applying        | 7     | 5.5        |
| Analysing       | 0     | 0          |
| Evaluating      | 0     | 0          |
| Creating        | 0     | 0          |
| <b>Total</b>    | 123   | 100        |